MAGNETS Spectroscopic Orchestration Layer

Infrastructure for Queue-Based LLAMAS Observing

Context. The MAGNETS collaboration pools Magellan time across partner institutions for Rubin transient follow-up. Per the Stubbs 2026B proposal: "We will apportion queue observing time and target prioritization in proportion to time awarded by respective TACs." This requires automated infrastructure to manage the queue schedule.

Problem. Current workflow (via Ashley's group) uses manual notebook-based scheduling. This doesn't scale with multiple requesters, lacks time accounting, and requires format translation between request intake and scheduling scripts.

Proposed Solution. An orchestration layer connecting RubinAlerts candidate discovery to LLAMAS observing plan generation:

Component	**Function**
Ingester	Pull targets from RubinAlerts `candidates.csv` and manual requests
Normalizer	Convert to common schema; estimate exposure times by redshift
Prioritizer	Score by merit × observability × moon phase compatibility
Time Accountant	Track 30-hour allocation across moon phases (5D + 20G + 5B)
LLAMAS Planner	Greedy scheduling with WD standard interleaving

LLAMAS-Specific Scheduling. From proposal Table 1, exposure times scale with redshift and moon phase:

Redshift	**Moon**	**Peak r**	**Exposure**	**Binning**
0.10–0.20	Bright	19.3	35 min	2×
0.20–0.30	Grey	20.6	95 min	2×
0.30–0.35	Grey/Dark	21.3	45–160 min	4×
0.35–0.40	Dark	21.9	180 min	4×

IFU advantages: ∼1 min overhead (vs. ∼10 min for slit), no slit losses, host galaxy "for free."

Integration with RubinAlerts. The existing pipeline produces nightly `candidates.csv` with merit-scored SNe Ia. The orchestration layer will:

- Filter to $z = 0.1$–0.4, $r = 18$–21.5, elliptical hosts preferred
- Map merit scores to LLAMAS priorities
- Generate queue-ready observing plans with WD standards (Boyd et al. 2026)

Implementation Timeline. Phase 1 (Wk 1–2): Normalizer, LLAMAS planner, CLI. Phase 2 (Wk 3): Time accounting, prioritizer. Phase 3 (Wk 4): RubinAlerts integration. Phase 4 (Wk 5+): WD standards, reporting.

Open Questions. (1) LLAMAS-specific constraints to confirm with Rob Simcoe; (2) Time charging: on schedule or observation? (3) Multi-night lookahead vs. nightly optimization?

Resources. Full design doc: `docs/design/spectroscopic-orchestration.md`